A System of Mutually Reliant Ratios
Creates the Internal Consistency of the System

Works in the Chinese li and the metric metre.

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Preface: Internal Consistency

> *"An ideal system of fundamental constants would be one in which these theoretical relationships were satisfied rigorously, while the adopted value of each individual constant agreed with its observed value."*
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> W. de Sitter & D. Brouwer (1938), *On the System of Astronomical Constants*, p. 213. De Sitter adds, p. 213: **"our present system is not consistent."**

A distance is never measured directly. It is computed from an angle and a known length, by trigonometry: a triangle is fixed only when **two of its three** parts are known. Feed in a radius R and an angle, and a distance comes out. Change the unit of R and every distance changes by the same factor. The unit is a free choice; the geometry is not. This brief runs the whole derivation in two units, the Chinese **li** and the metric **metre**, and shows the two results convert exactly into each other.

Part I: The Chinese System

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1 The angle: the du

The Chinese divided one full turn of the sky into **365.25 du**, one du for each day the Sun drifts through the stars in a year. One du is therefore just under one degree. Each du holds 10 fen.

$$1 \text{ du} = \frac{360°}{365.25} = 0.9856° = 59.14'$$

2 The length: the li

The li is the Chinese unit of ground distance (1 li = 300 bu). In 724 AD the astronomer Yi Xing tied it to the sky: marching north along a ground meridian, the pole star climbs **1 du for every 351.267 li** (Beer et al., 1961). Closed into a full circle of 365.25 du, the survey gives the circumference of the heavens in li, and the radius follows from $C = 2\pi R$.

$$\text{celestial circle} = 351.267 \times 365.25 \approx 128{,}300 \text{ li} \qquad R = \frac{\text{circle}}{2\pi} \approx 20{,}420 \text{ li}$$

Gai Tian cosmology. This is the Chinese canopy-heaven model, *Gai Tian* (Cullen, 1996; Needham, 1959): the ground is a flat plane and the li measures travel across it; the du divides the sky, the Heavens turning around the observer. The circle and radius R above are dimensions of the celestial sphere, not of a terrestrial globe. The arc is in the sky, a matter of optics; the ground itself does not curve. The frame is observer-centric, each observer at the centre of his own sky.

3 The Moon and the Sun, in li

Parallax turns an angle into a distance. A body at distance d, seen against the radius R from Section 2, subtends an angle θ fixed by the relation below; supply R and that angle and the distance follows. Both angles used here are parallax angles. θ_{moon} is the Moon's horizontal parallax, the shift in its apparent place seen across a baseline R. θ_{sun} is the solar parallax, fixed from timing a transit of Venus or Mercury across the Sun's face.

$$d = \frac{R}{\sin\theta}$$

$$\text{Moon } (\theta_{\text{moon}} = 57') : \quad d = \frac{20{,}420}{\sin 57'} \approx 1{,}231{,}600 \text{ li}$$

$$\text{Sun } (\theta_{\text{sun}} = 8.79'') : \quad d = \frac{20{,}420}{\sin 8.79''} \approx 4.789 \times 10^{8} \text{ li} \ = \ 1 \text{ AU}$$

Kepler's third law gives each planet's orbit as a fixed ratio to Earth's. Its distance is that ratio times the AU.

Planet	**Orbit ratio**	**Distance (li)**
Mercury	0.387	1.854×10^{8}
Venus	0.723	3.464×10^{8}
Earth	1.000	4.789×10^{8}
Mars	1.524	7.298×10^{8}
Jupiter	5.203	2.492×10^{9}
Saturn	9.555	4.576×10^{9}
Uranus	19.218	9.205×10^{9}
Neptune	30.110	1.442×10^{10}

4 The speed of light and aberration, in li

The speed of light is a property of the vacuum, $c = 1/\sqrt{\mu_0\,\varepsilon_0}$. Measured against li-based units it comes to about 959,787 li per second.

$$c = \frac{1}{\sqrt{\mu_0\,\varepsilon_0}} \approx 959{,}787 \text{ li/s}$$

Aberration route. Earth's assumed motion is said to tilt starlight by the aberration angle κ (Bradley, 1728), with $\tan\kappa = v_\oplus/c$ and $v_\oplus = 2\pi\,\text{AU}/T$, where T is the year. Solved for the AU, this returns the Earth–Sun distance from c and κ alone, the route Foucault ran in 1862 from a measured c and Bradley's κ (Foucault, 1862).

$$v_\oplus = c\,\tan\kappa \approx 95.37\ \text{li/s} \qquad 1\ \text{AU} = \frac{v_\oplus T}{2\pi} \approx 4.790\times 10^8\ \text{li}$$

$$\kappa + N + p = 20.50'' + 9.20'' + 50.29'' \approx 79.99''$$

The aberration angle is fixed, not measured. A telescope records one quantity: the total stellar oscillation, the trepidation angle $\kappa + N + p \approx 80''$. Nothing in the observation splits it into aberration, nutation and precession. The route needs κ to be exactly the slice for which $c\,\tan\kappa$ equals the Kepler velocity $2\pi\,\text{AU}/T$; that slice is not read off the oscillation, it is assigned the value that closes the identity. In 1805 Laplace had Delambre redetermine the light constant from the eclipses of Jupiter's moon Io, and Delambre returned exactly Bradley's aberration figure (Laplace, 1805). The aberration angle κ and the solar parallax $\pi_\odot$ are both pinned to the AU, so the aberration route and the parallax route of Section 3 return the same distance: that agreement is built in, not independent confirmation. In the observer-centric Gai Tian frame the Earth does not move. The telescope records one angle, the whole oscillation $\kappa + N + p$; the velocity read from it, $c\,\tan(\kappa + N + p)$, is the speed of nothing.

Part II: The Metric System

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5 The angle and the metre

The metric angle is the **degree**: one full turn is 360 degrees.

The **metre** began in 1791 as 1/10,000,000 of the Earth's meridian quadrant, pole to equator, a length pegged to the pole star as the li is. The Earth is an oblate spheroid; taking its equatorial circumference, 40,075 km, the equatorial radius follows.

$$\text{equatorial circumference} = 40{,}075\ \text{km} \qquad R = \frac{40{,}075\ \text{km}}{2\pi} \approx 6{,}378\ \text{km}$$

On the radius. The Earth is an oblate spheroid: its equatorial radius $a = 6{,}378.137$ km exceeds its polar radius $b = 6{,}356.752$ km. This brief uses the equatorial radius. The commonly quoted 6,371 km is the IUGG mean radius $R = (2a + b)/3$, an average over the spheroid; it is not used here.

The two-sphere model. The globe account drops the observer-centric sky for a geocentric one. It places a fictitious observer at the centre of the Earth; this linearises the separate observer hemispheres into one sphere and sets a terrestrial sphere and the celestial sphere in 1:1 correspondence. The radius R here is that terrestrial sphere. Star positions and ground positions are both referred to the single fictitious centre.

6 The Moon and the Sun, in metres

The same parallax law $d = R/\sin\theta$, the same observed angles, the metric radius from Section 5.

$$\text{Moon } (\theta = 57') : \quad d = \frac{6{,}378 \text{ km}}{\sin 57'} \approx 384{,}690 \text{ km}$$

$$\text{Sun } (\theta = 8.79'') : \quad d = \frac{6{,}378 \text{ km}}{\sin 8.79''} \approx 1.496 \times 10^8 \text{ km} \ = \ 1 \text{ AU}$$

Planet	**Orbit ratio**	**Distance (km)**
Mercury	0.387	5.791×10^7
Venus	0.723	1.082×10^8
Earth	1.000	1.496×10^8
Mars	1.524	2.279×10^8
Jupiter	5.203	7.783×10^8
Saturn	9.555	1.429×10^9
Uranus	19.218	2.875×10^9
Neptune	30.110	4.505×10^9

7 The speed of light and aberration, in metres

The same vacuum relation, in metric units. Because μ_0 and ε_0 are measured **per metre**, $c = 1/\sqrt{\mu_0 \, \varepsilon_0}$ comes out in metres per second.

$$c = \frac{1}{\sqrt{\mu_0 \, \varepsilon_0}} = 299{,}792{,}458 \text{ m/s}$$

The aberration route runs as in Section 4, with c in m/s and T the year:

$$v_\oplus = c \tan\kappa \approx 29{,}789 \text{ m/s} \qquad 1 \text{ AU} = \frac{v_\oplus T}{2\pi} \approx 1.496 \times 10^8 \text{ km}$$

8 Light-time of flight, in metres

A distance can be restated as the time light takes to cross it. One light-second is $c \times 1$ s; one light-minute is sixty of those. Dividing any distance by c gives its light-time.

$$1 \text{ light-second} \approx 299{,}792 \text{ km} \qquad 1 \text{ light-minute} \approx 1.799 \times 10^7 \text{ km}$$

$$1 \text{ AU} \approx 8.32 \text{ light-minutes}$$

Planet	**Light-time**	**Planet**	**Light-time**
Mercury	3.22 min	Jupiter	43.27 min
Venus	6.02 min	Saturn	79.47 min
Earth	8.32 min	Uranus	159.8 min
Mars	12.67 min	Neptune	250.4 min

9 The metre and c: an algebra switch

Until 1983 the metre was the standard and c was measured against it. In 1983 c was fixed at exactly 299,792,458 m/s and the metre was redefined as the distance light travels in $1/c$ seconds.

$$\frac{1}{c} = 3.3356 \text{ ns} \qquad \text{metre} = c \times \frac{1}{c} = 1 \text{ m}$$

Light crosses one metre in 3.3356 ns; multiply that time back by c and one metre returns, unchanged. Only the quantity held fixed changed. The length did not.

Part III: Conversion and Equivalence

10 Converting li to metres, and the summary

The li circle measures the heavens; the metric circle measures the terrestrial sphere. The two-sphere model sets the two in 1:1 correspondence, so the celestial circumference 128,300 li and Earth's circumference 40,075 km describe one circle. Dividing gives the conversion factor.

$$1 \text{ li} = \frac{40{,}075{,}000 \text{ m}}{128{,}300 \text{ li}} \approx 312.35 \text{ m} = 0.31235 \text{ km}$$

Every distance is R times a factor built only from angles. The angle factors are identical in both systems, so multiplying any li value by 312.35 m/li reproduces the metre value exactly.

Quantity	**Chinese (li)**	**Metric (m / km)**
Radius R	20,420 li	6,378 km
Moon distance	1.232×10^6 li	384,690 km
Sun distance (1 AU)	4.789×10^8 li	1.496×10^8 km
AU via aberration	4.790×10^8 li	1.496×10^8 km
1 light-minute	5.759×10^7 li	1.799×10^7 km
Speed of light c	959,787 li/s	299,792,458 m/s

Check. 1,231,591 li × 312.35 m/li = 384,691,000 m = 384,691 km (the Moon, both ways)

The li ladder and the metric ladder are one ladder, read through the 312.35 factor. Neither unit is more correct than the other.

Closing

Each distance here is fixed by one chosen radius and a set of measured angles. Run the calculation in li or in metres and only the labels change; convert between them and every value matches. A system of mutually reliant ratios is internally consistent by construction. As de Sitter noted, that is not the same as being independently verified.

References

Beer, A., Ho Ping-Yü, Lu Gwei-Djen, Needham, J., Pulleyblank, E. G., & Thompson, G. I. (1961). An 8th-century meridian line: I-Hsing's chain of gnomons and the pre-history of the metric system. *Vistas in Astronomy*, 4, 3–28.

Bradley, J. (1728). An account of a new discovered motion of the fixed stars. *Philosophical Transactions of the Royal Society*, 35, 637–661.

Cullen, C. (1996). *Astronomy and Mathematics in Ancient China: The Zhou Bi Suan Jing.* Cambridge: Cambridge University Press.

de Sitter, W., & Brouwer, D. (1938). On the system of astronomical constants. *Bulletin of the Astronomical Institutes of the Netherlands*, 8(307), 213–231.

Foucault, L. (1862). Détermination expérimentale de la vitesse de la lumière; parallaxe du Soleil. *Comptes Rendus de l'Académie des Sciences*, 55, 501–503.

Laplace, P.-S. (1805). *Traité de Mécanique Céleste, Tome IV: Théorie des satellites de Jupiter*. Paris: Courcier.

Needham, J. (1959). *Science and Civilisation in China, Volume 3: Mathematics and the Sciences of the Heavens and the Earth*. Cambridge: Cambridge University Press.